

Deterministic Optimization

SOC, SDP Examples

Andy Sun

Assistant Professor

Stewart School of Industrial and Systems Engineering

Extremal Ellipsoid Problem

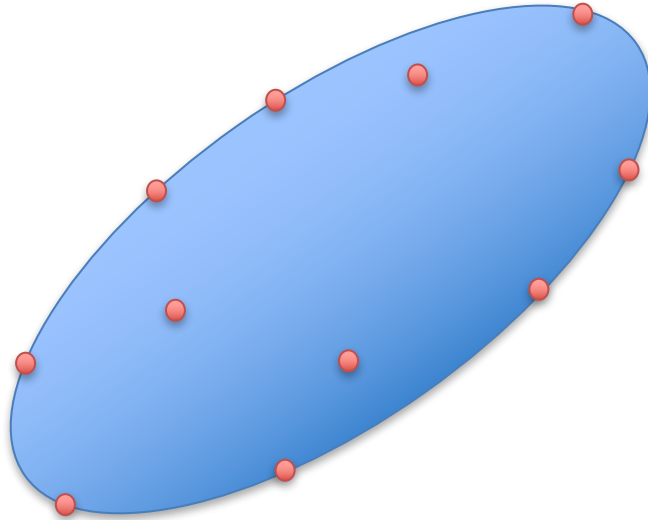
SOCP, SDP Examples

Learning Objectives for this lesson

- Learn about an extreme ellipsoid problem
- Discover an SOCP-SDP formulation

An Extremal Ellipsoidal Problem

- Suppose $D \subseteq \mathbb{R}^n$ is a set of data points $D = \{v_1, \dots, v_m\}$
- We want to find a “smallest” ellipsoid that contains D



How to Describe an Ellipsoid

- Describe an ellipsoid E centered at c , with shape matrix Q
 - $E := \{x \in \mathbb{R}^n: (x - c)^\top Q (x - c) \leq 1\}$
 - $Q \succcurlyeq 0$
- Different ways to measure the “size” of an ellipsoid
- A (most) common way is to use the volume of an ellipsoid
- The volume of the above ellipsoid is proportional to
 - $\text{Vol}(E) \sim \det Q^{-1}$
 - Moreover, $f(Q) := \log \det(Q^{-1})$ is a **convex function** of Q !

Minimum Volume Circumscribing Ellipsoid

- Find an ellipsoid E with minimum volume that contains D
- $\min \log \det Q^{-1}$
- S.t. $(v_i - c)^\top Q (v_i - c) \leq 1 \quad \forall i = 1, \dots, m$
- However, this is not a convex problem
- We need to do some transformation:
 - Let $Q = AA^\top$, so $A = Q^{1/2} \rightarrow \det Q^{-1} = \det(A^{-2}) = (\det(A^{-1}))^2$
 - $\rightarrow \log \det Q^{-1} = 2 \log \det A^{-1}$
 - $(v_i - c)^\top Q (v_i - c) \leq 1 \rightarrow \|Av_i - Ac\|_2 \leq 1 \rightarrow \|Av_i - b\|_2 \leq 1$

Convex Reformulation

- An SOCP, SDP reformulation
- $\min \log \det A^{-1}$
s.t. $\|Av_i - b\|_2 \leq 1 \quad \forall i = 1, \dots, m$
 $A \succcurlyeq 0$
- Equivalent to
- $\min \log \det A^{-1}$
s.t. $\begin{bmatrix} Av_i - b \\ 1 \end{bmatrix} \in L_{n+1} \quad \forall i = 1, \dots, m,$
 $A \succcurlyeq 0$

Remember L_{n+1} is the $n+1$
Dimensional second-order cone:

$$\left\{ (x_1, \dots, x_n, x_{n+1}) : \sum_{i=1}^n x_i^2 \leq x_{n+1}^2 \right\}$$

Summary

- We learned:
 - What is the minimum volume circumscribing ellipsoidal problem
 - How to formulate an SOCP-SDP reformulation